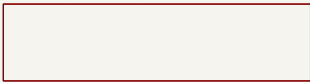


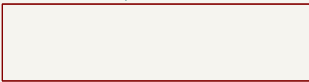
UNDER-DASH POWER BOARD: block diagram. Each box is a sheet (double-click it in KiCad). Power rails +12V, +5V, +3V3 and GND join the sheets by name (power symbols); every other connection between sheets is a wire here.

12 V input and protection



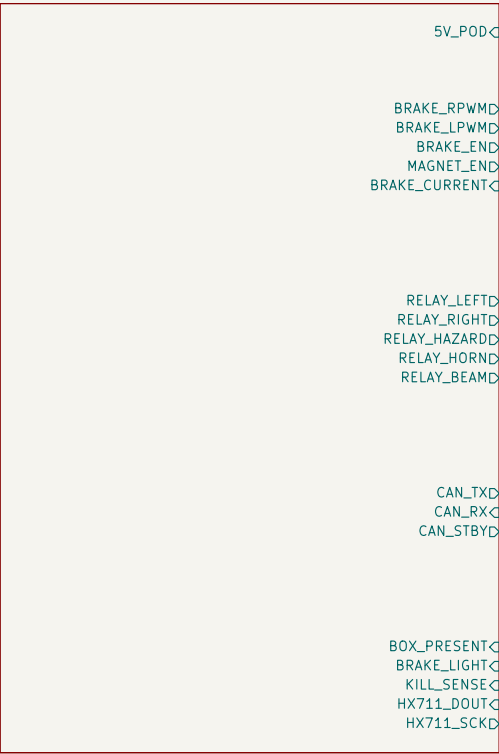
File: input.kicad\_sch

5 V converter (TPS54360, TI reference design)



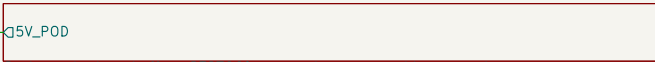
File: buck.kicad\_sch

Inter-box cable: DB-25 to the pod

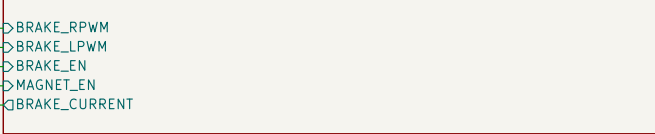


File: cable.kicad\_sch

3.3 V, the pod's 5 V, the LIDAR feed

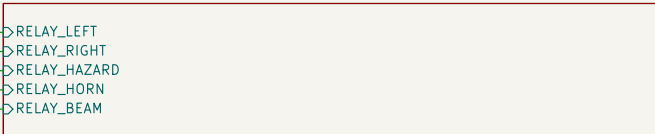


Brake, actuator, buffer, BTS7960, current sense, cable magnet



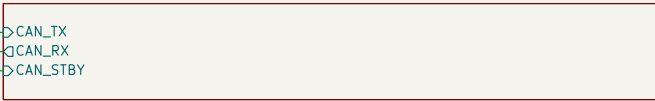
File: actuator.kicad\_sch

Signal relays: indicators, hazards, horn, high beam



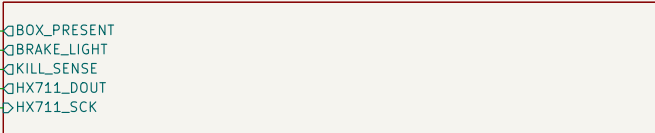
File: relays.kicad\_sch

CAN transceiver for OBD-II



File: can.kicad\_sch

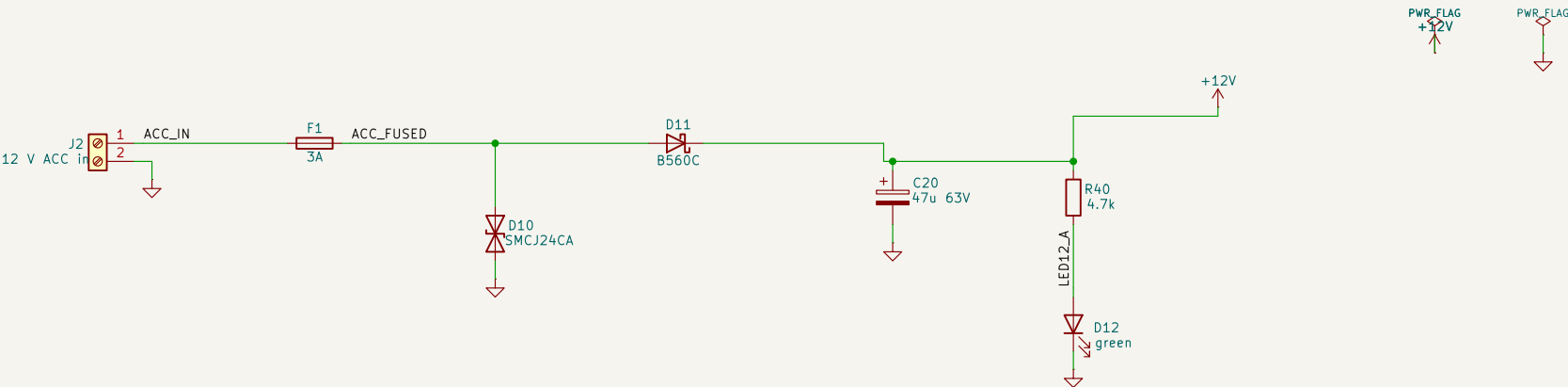
Presence, brake light, E-stop sense, kill relay, bench



File: status.kicad\_sch

12 V input -> 5 V converter -> 3.3 V and the pod's 5 V.  
The pod drives the actuator, relays and CAN;  
the status sheet reports back to it.

12 V INPUT (the logic feed): switched ACC from the car's fuse box, so the board is off with the ignition. F1 3 A fuse. D10 bidirectional TVS clamps load-dump and jump-start spikes to about 39 V. D11 Schottky (60 V, 5 A) blocks a reversed lead. D12: +12V present.



Mounting holes: M3, no electrical connection

H1  
M3

H2  
M3

H3  
M3

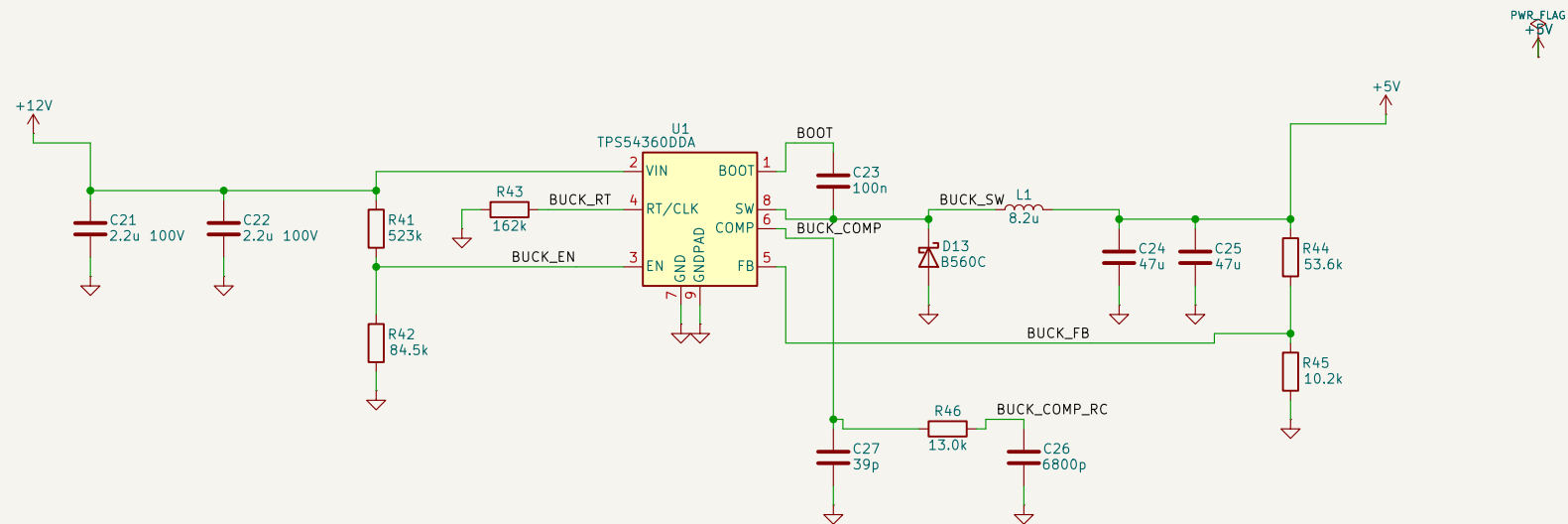
H4  
M3

Generated by hardware/gen; see BUILD\_GUIDE Part 4.8  
Under-dash power board (power, actuator, relays, CAN)  
**AR Windscreen FYP – Kiogora Ian Mwenda**  
Sheet: /12 V input and protection/  
File: input.kicad\_sch

**Title: 12 V input and protection**

|                     |                  |         |
|---------------------|------------------|---------|
| Size: A4            | Date: 2026-10-07 | Rev: 1  |
| KiCad E.D.A. 10.0.1 |                  | Id: 2/9 |

5 V: TI's TPS54360 5 V / 3.5 A reference design (datasheet SLVSBB4G, Figure 34), unchanged.  
Rated to 60 V input, so whatever the TVS lets through a load dump cannot destroy it.  
R41/R42 start it at 8 V and stop it at 6.25 V. R43 sets 600 kHz. R44/R45 set 5.0 V.  
C23 is the bootstrap for the high-side switch; D13 carries the current while it is off.  
R46, C26, C27 compensate the control loop. Lay out C21/C22, U1, D13 and L1 as one tight loop.



Generated by hardware/gen; see BUILD\_GUIDE Part 4.8  
Under-dash power board (power, actuator, relays, CAN)

AR Windscreen FYP – Kiogora Ian Mwenda

Sheet: /5 V converter (TPS54360, TI reference design)/  
File: buck.kicad\_sch

Title: 5 V converter (TPS54360, TI reference design)

Size: A4

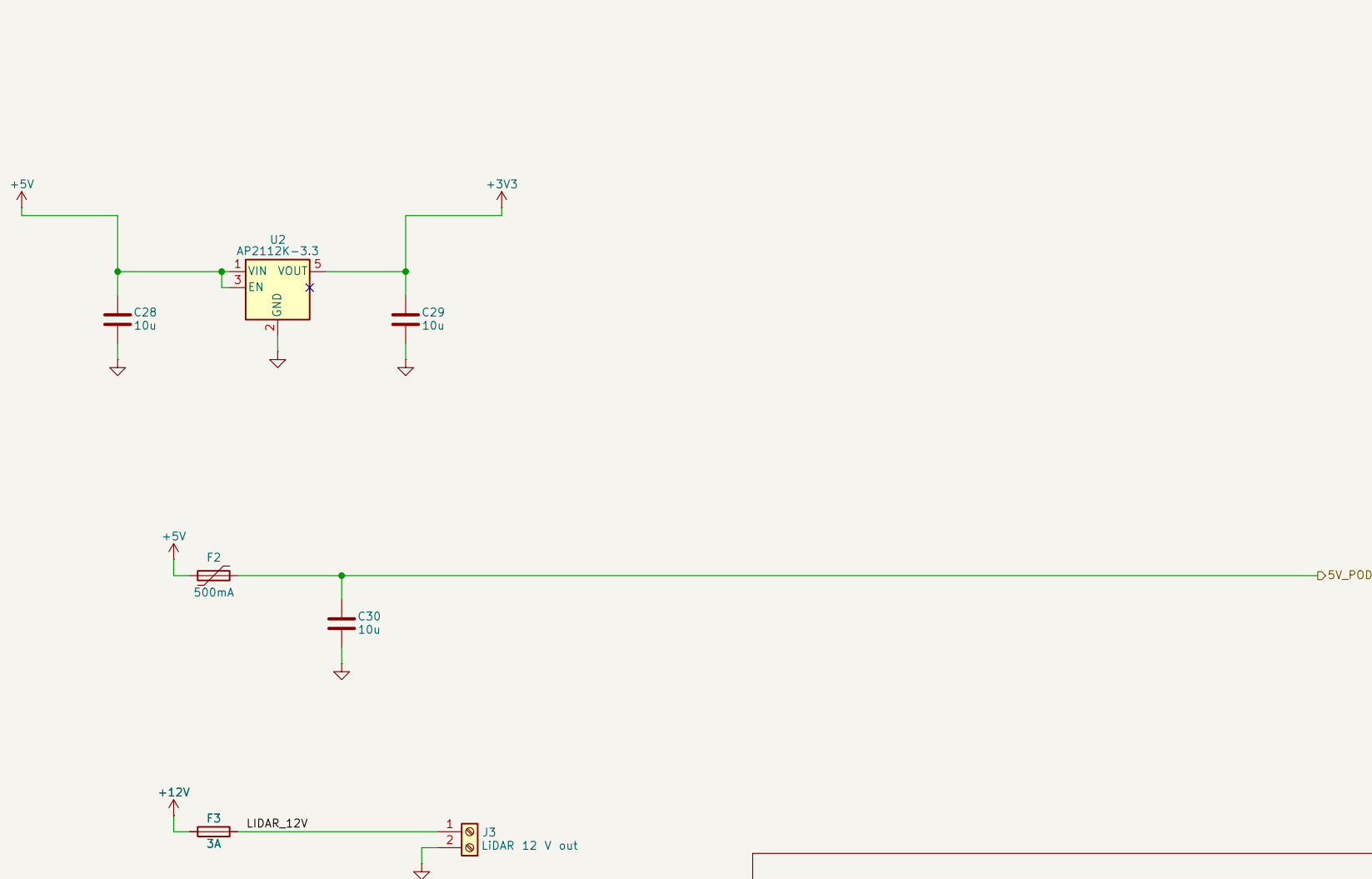
Date: 2026-10-07

Rev: 1

KiCad E.D.A. 10.0.1

Id: 3/9

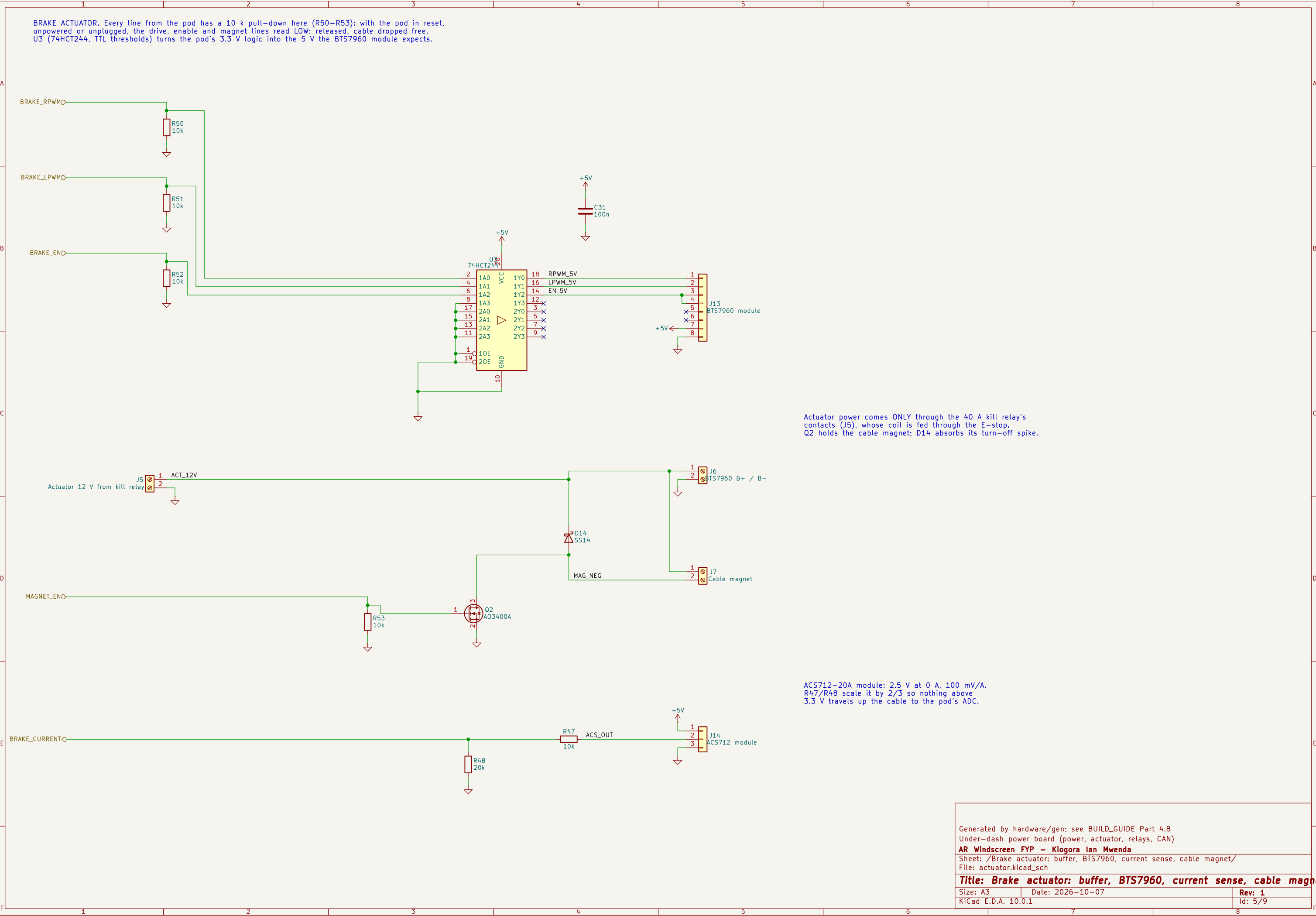
3.3 V (AP2112K) for the CAN transceiver only; the pod makes its own 3.3 V.  
5V\_POD: the pod's supply up the cable, through F2, a 500 mA resettable fuse,  
so a short in the cable cannot take down this board's 5 V.  
LiDAR: 12 V through its own 3 A fuse F3 to the M12 socket.



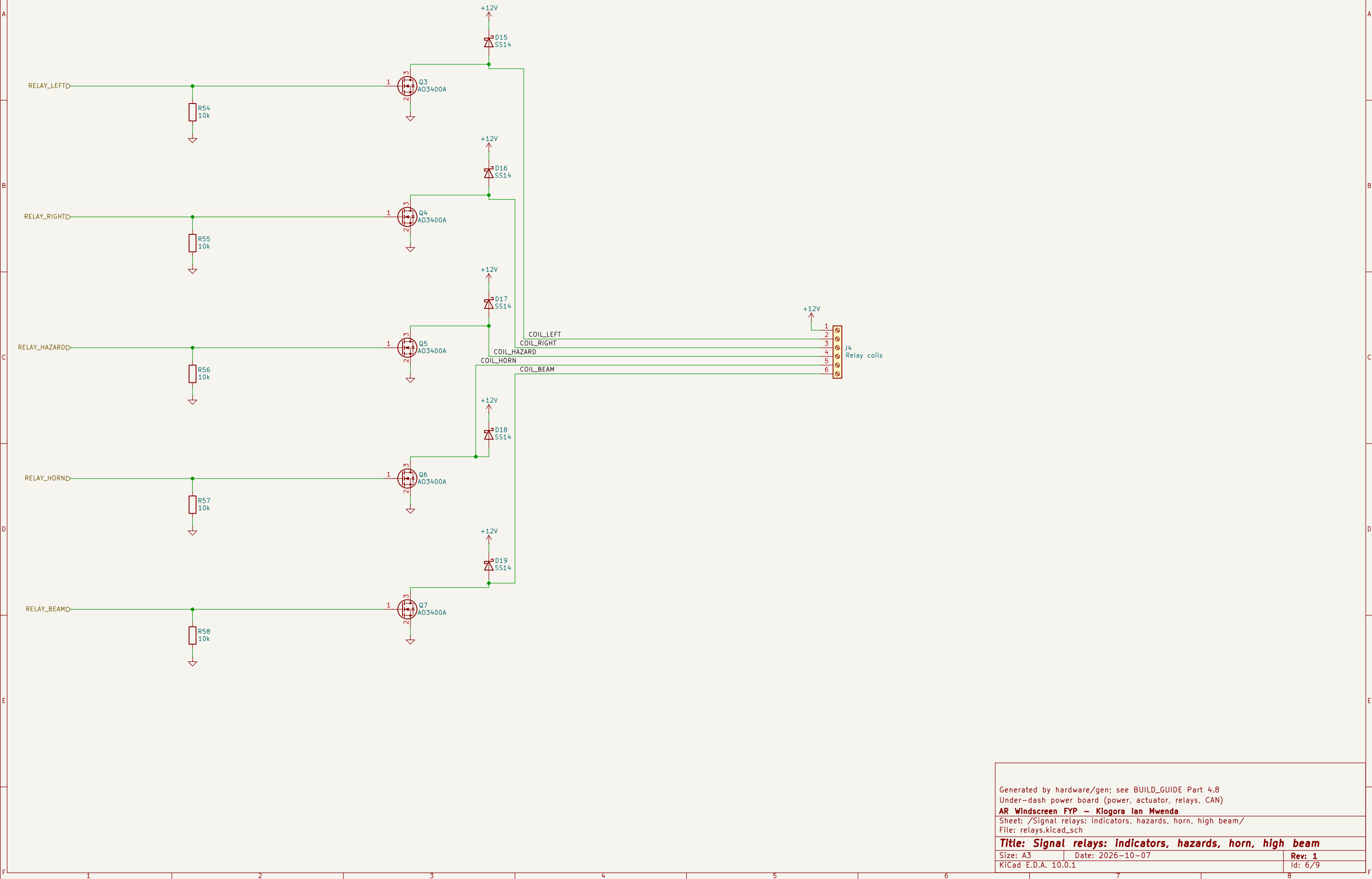
Generated by hardware/gen; see BUILD\_GUIDE Part 4.8  
Under-dash power board (power, actuator, relays, CAN)  
**AR Windscreen FYP – Kiogora Ian Mwenda**  
Sheet: /3.3 V, the pod's 5 V, the LiDAR feed/  
File: rails.kicad\_sch

**Title: 3.3 V, the pod's 5 V, the LiDAR feed**

|                     |                  |         |
|---------------------|------------------|---------|
| Size: A4            | Date: 2026-10-07 | Rev: 1  |
| KiCad E.D.A. 10.0.1 |                  | Id: 4/9 |



SIGNAL RELAYS: five 12 V automotive relays in sockets, off the board; their contacts tap the car's switch wiring (Part 15.2). Each coil is switched low-side by a MOSFET whose gate has a 10 k pull-down (R54–R58): no drive from the pod, no relay. D15–D19 absorb each coil's turn-off spike.

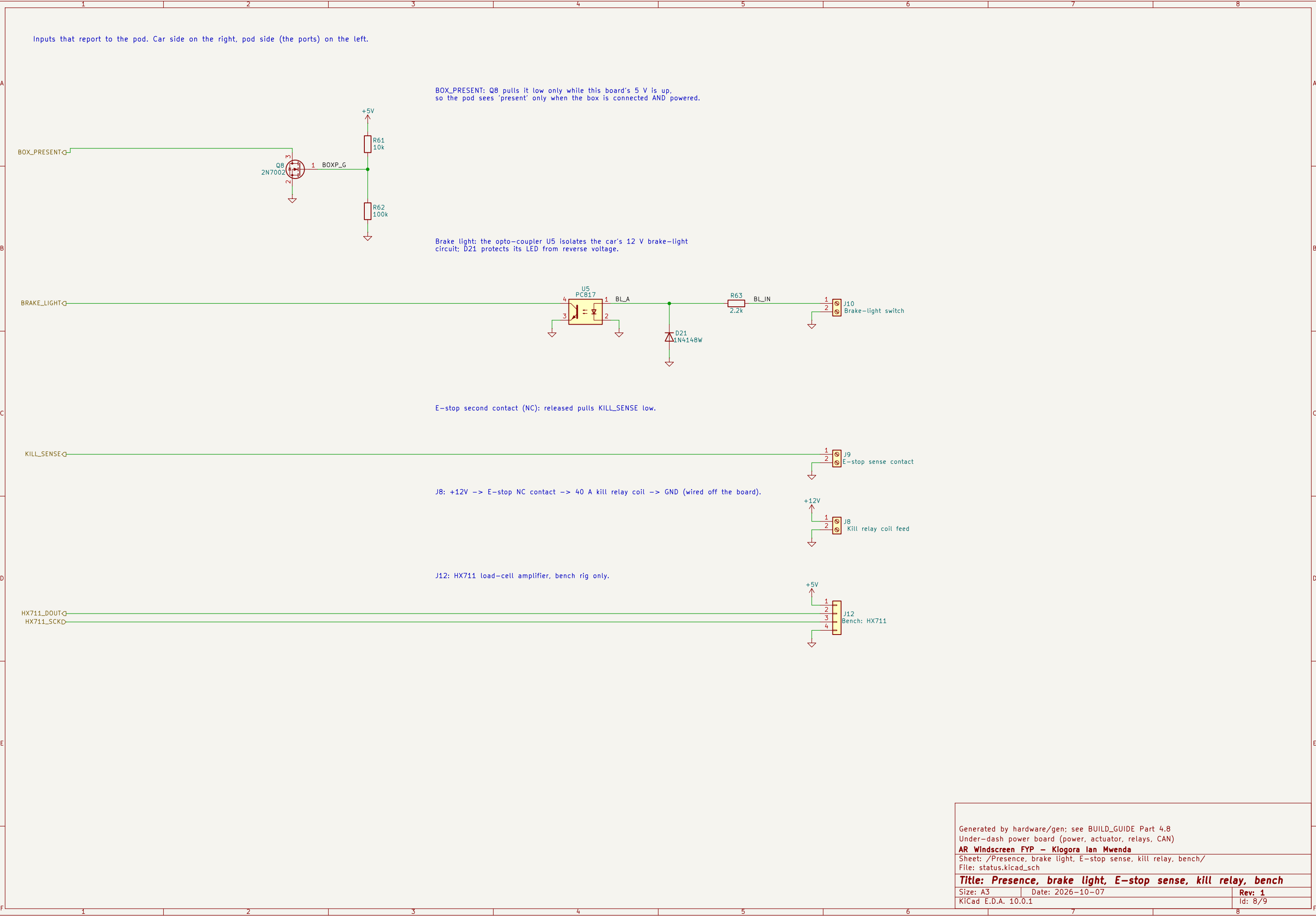


CAN (ISO 15765-4) to the car's OBD-II port, on this board so the unterminated branch off the car's bus is only the ~0.5 m OBD lead. R59 holds TX recessive and R60 holds Rs high (standby, cannot transmit) whenever the pod is not driving them. No termination: the car has its two.

Generated by hardware/gen; see BUILD\_GUIDE Part 4.8  
Under-dash power board (power, actuator, relays, CAN)  
**AR Windscreen FYP – Klogora Ian Mwenda**  
Sheet: /CAN transceiver for OBD-II/  
File: can.kicad\_sch

|                                          |                  |         |
|------------------------------------------|------------------|---------|
| <b>Title: CAN transceiver for OBD-II</b> |                  |         |
| Size: A4                                 | Date: 2026-10-07 | Rev: 1  |
| KiCad E.D.A. 10.0.1                      |                  | Id: 7/9 |

Id: 7/9



Generated by hardware/gen; see BUILD\_GUIDE Part 4.8  
Under-dash power board (power, actuator, relays, CAN)

AR Windscreen FYP - Klogora Ian Mwenda  
Sheet: /Presence, brake light, E-stop sense, kill relay, bench/  
File: status.kicad\_sch

Title: Presence, brake light, E-stop sense, kill relay, bench

|                     |                  |         |
|---------------------|------------------|---------|
| Size: A3            | Date: 2026-10-07 | Rev: 1  |
| KiCad E.D.A. 10.0.1 |                  | Id: 8/9 |



1 2 3 4 5

A

B

C

D

1 2 3 4 5

INTER-BOX CABLE (BUILD\_GUIDE 4.8.3). Shield bonded to GND at this end only.  
Every line driven by the pod has its fail-safe pull-down on the sheet that uses it.

J1  
DB-25 to pod

SH

1 14 2 15 3 16 4 17 5 18 6 19 7 20 8 21 9 22 10 23 11 24 12 25 13

5V\_POD  
RELAY\_LEFT  
RELAY\_RIGHT  
RELAY\_HAZARD  
RELAY\_HORN  
BRAKE\_RPWM  
RELAY\_BEAM  
BRAKE\_LIGHT  
BRAKE\_LPWM  
CAN\_TX  
BRAKE\_EN  
CAN\_RX  
MAGNET\_EN  
CAN\_STBY  
BRAKE\_CURRENT  
HX711\_DOUT  
HX711\_SCK  
KILL\_SENSE  
BOX\_PRESENT

Generated by hardware/gen; see BUILD\_GUIDE Part 4.8  
Under-dash power board (power, actuator, relays, CAN)  
**AR Windscreen FYP – Klogora Ian Mwenda**  
Sheet: /Inter-box cable: DB-25 to the pod/  
File: cable.kicad\_sch

**Title: Inter-box cable: DB-25 to the pod**

|                     |                  |        |
|---------------------|------------------|--------|
| Size: A4            | Date: 2026-10-07 | Rev: 1 |
| KiCad E.D.A. 10.0.1 | Id: 9/9          |        |

Id: 9/9